

Size and level of a test example

Calculating size of a test

Let $X_1, \dots, X_{100} | \theta \stackrel{iid}{\sim} \text{Bern}(\theta)$ where $\theta \in [0, 1]$.

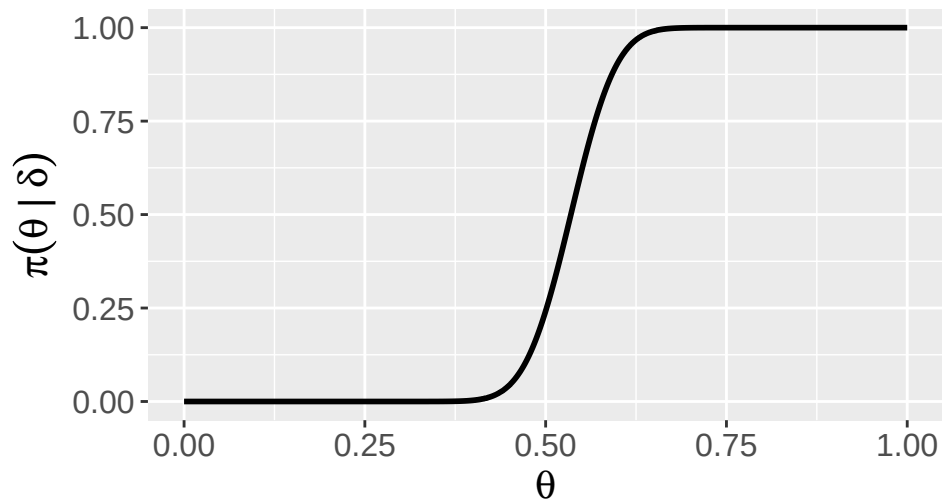
$$H_0 : \theta \leq 0.5 \qquad H_1 : \theta > 0.5$$

For our testing procedure δ , define $r(\mathbf{X}) = \sum_{i=1}^{100} X_i$ and $R = \{r(\mathbf{X}) \geq 54\}$.

Visualize power function

We can see that the power function is increasing in θ :

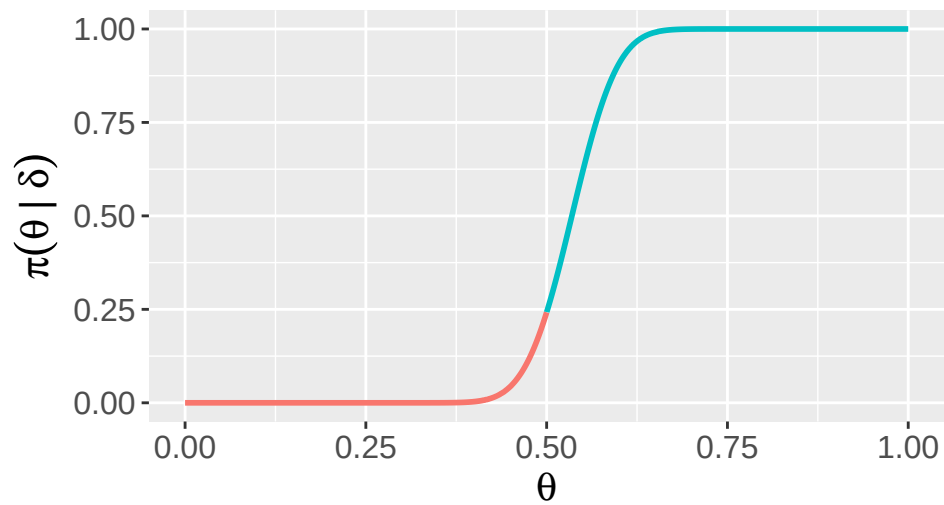
$$\pi(\theta | \delta) = \Pr \left(\sum_{i=1}^{100} X_i \geq 54 | \theta \right) = \sum_{k=54}^{100} \binom{100}{k} \theta^k (1-\theta)^{100-k}$$



Find size of test

How to find its size? Recall:

$$\alpha(\delta) = \sup_{\theta \in \Omega_0} \pi(\theta | \delta)$$



We see that the sup occurs at $\theta = 0.5$! So:

$$\alpha(\delta) = \sup_{\theta \in \Omega_0} \Pr \left(\sum_{i=1}^{100} X_i \geq 54 | \theta \right) = \Pr \left(\sum_{i=1}^{100} X_i \geq 54 | \theta = 0.5 \right)$$

Using R, the size of our test is:

```
# pbinom(x, n, p) = P(X <= x)
1 - pbinom(53, n, 0.5)
```

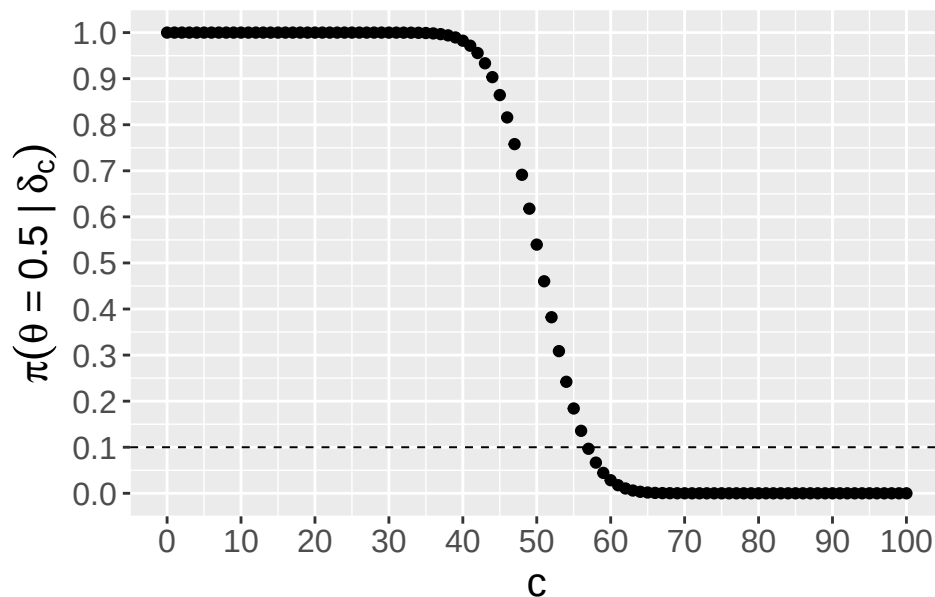
```
[1] 0.2420592
```

Finding a level- α_0 test

Brute-force approach

Now suppose I consider tests δ_c that have rejection regions $R = \{r(\mathbf{X}) \geq c\}$ for some c , where $r(\mathbf{X}) = \sum_{i=1}^{100} X_i$ as before. I want to find a c that yields a level-10 test.

The following plots $\Pr(\sum X_i \geq c | \theta = 0.5)$ for $c = 0, 1, \dots, 100$. How can we use this to determine which c gives us a level-0.10 test?



We can see that these probabilities drop below 0.10 when $c = 57$! So the testing procedure that has rejection region $R = \{r(\mathbf{X}) \geq 57\}$ is a level-0.10 test of these hypotheses!

```
# Confirm: P(sum Xi >= 57 | theta = 0.5) <= 0.10  
1 - pbinom(57 - 1, n, 0.5)
```

```
[1] 0.09667395
```

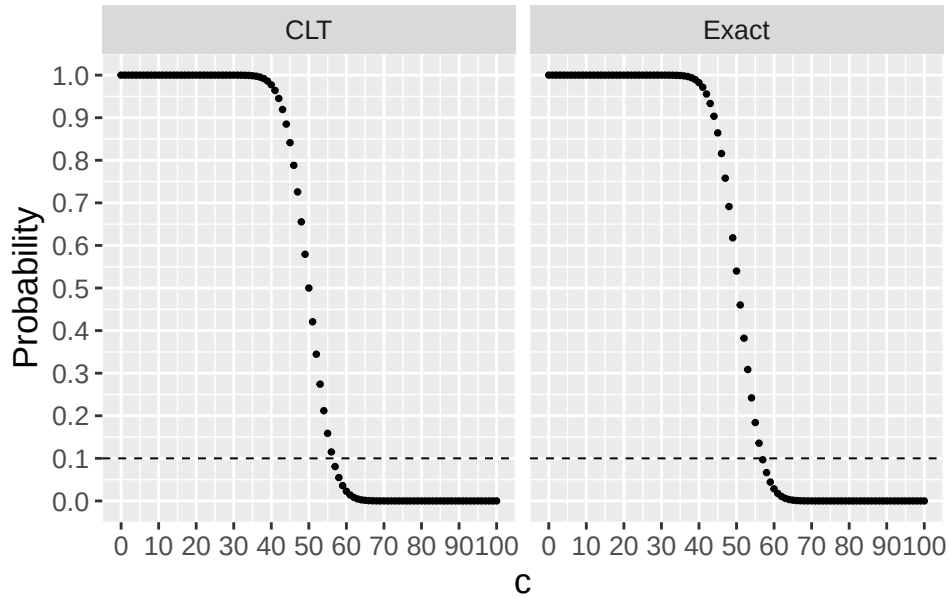
Question: does this choice of $c = 57$ yield the only level-0.10 test?

CLT approximation

Using the CLT approach, we found

$$\Pr \left(\sum_{i=1}^{100} X_i \geq c \mid \theta = 0.5 \right) \approx 1 - \Phi \left(\frac{c - 50}{5} \right)$$

We will also plot this as a function of c to find a level-0.10 test, next to the plot from before with the exact probabilities shown on the previous plot. Note that they are quite similar!



We can see that the CLT yields the same choice of $c = 57$.

Let's obtain the CLT approximated probability:

```
# verify CLT approximation  
1 - pnorm( (57 - 50) / 5)
```

```
[1] 0.08075666
```